



GDCAU

Green Deal for Central Asian Universities

Project 101179245

The Regenerative Cycle: From Observation to Action

What does it really mean to act on sustainability? For university students in Central Asia and beyond, it's not just about studying climate science or environmental policy—it's about engaging directly with the systems we live in and helping to repair and renew them. It is to ask not only how to reduce harm, *but how to create conditions where both people and nature can thrive*. As Buckton et al. describe it, regenerative systems enable “life to beget life”—they restore soils, build community, strengthen resilience, and spark creativity. This approach transforms learning: instead of absorbing knowledge from lectures alone, students become co-creators of change through local, practical action. From restoring a polluted stream to rethinking how food is grown, regeneration invites us to observe, reflect, design, and act in ways that reconnect us to place and each other.

This guide introduces a **five-step Regenerative Cycle**—Observe → Connect → Identify → Design → Learn—that helps you move from ideas to action. Rooted in systems thinking and experiential learning, it offers simple tools, examples, and practical resources to help you understand local issues, find leverage points, and co-design practical solutions. Whether you are studying environmental science, economics, or education, this cycle supports hands-on learning and leadership for a healthier campus and more regenerative future.

1. Observe

Start by *listening* to the place. Walk around your campus or neighborhood with curiosity. Note *ecological patterns* (wind, sun, water flow, plants, animals) **and** *social patterns* (people's schedules, cultural gatherings, commerce). For example, you might observe how

winter snows melt into an irrigation canal or see where cafeteria scraps are thrown out. Look with all your senses and from different angles. Carry a notebook or use your phone to sketch or record what you see and hear.

Reflection prompts – After observing:

- What did you notice about natural elements (soil, water, sunlight, living things)? What about people's activities?
- Is there anything in the landscape that seems *out of place* (like a broken gutter or bare patches)? Why might that be?
- Interview a local expert (e.g. a farmer, gardener, elder): what changes have they seen in the land or community? What questions do they suggest you explore?

Examples and tools:

- **Nature Positive Universities** encourages students to conduct biodiversity audits of their campuses (trees, pollinators, birds, etc.) as a first step toward nature-positive planning. <https://www.naturepositiveuniversities.net/>
- **Re:wild Your Campus** has a student toolkit for doing environmental impact assessments (soil quality, pesticide use, lawn monocultures). <https://www.rewildyourcampus.org/>
- Students at the **University of Sussex (UK)** created visual maps of where biodiversity was missing or under threat on their campus to inspire design ideas.

2. Connect

Next, map out how things are linked. Everything you observed is part of a **system**. Draw simple diagrams or talk with friends about how water, food, energy, waste and social life in your area form a network. For instance, if your university uses a lot of water for landscaping, how does that water reach your campus? Where does used water go? How might wasted food from the cafeteria end up affecting local rivers or gardens? In Central Asia's mountains, melting snow feeds rivers that provide irrigation and hydropower; a drought can ripple through both farms and electricity supplies.

Use *systems thinking* to spot these relationships. (See our guide on “Systems Thinking: A Beginner's Toolkit” for more information.) For example, consider **feedback loops**: watering a garden might improve soil (greenhouse gas sink) which cools local air (less energy for fans), etc. Try the iceberg model: what “events” did you see, what underlying patterns or causes connect them (e.g. culture, infrastructure, policies)?

Reflection prompts – Making connections:

- Draw a simple “web”: place observed elements (water source, garden, kitchen, electricity, people) and draw arrows showing how one influences another. Where do you see loops or cycles?
- Identify a *leverage point*: a small factor that could change many things. (E.g. planting trees could cool buildings, reduce energy use, prevent erosion.) Which factor seems most powerful?

- List any surprises: did observing one element make you think differently about another (for example, noticing trash made you think about consumption)? What new questions emerged?

Examples and tools:

- Use the **Second Nature Climate Justice Toolkit** to explore how climate challenges intersect with social inequalities (e.g. Which neighborhoods lack trees? Who is affected by poor air quality on campus?). <https://secondnature.org/resources/>
- **Circle U. University Alliance** recommends that students link environmental questions to social well-being, policy, and curriculum opportunities through “whole-university approaches”. <https://www.circle-u.eu/>

3. Identify

From your connections, pick **promising spots for change**. These are your leverage points and regenerative opportunities. For example, maybe you saw that the cafeteria’s food scraps are discarded but could become compost to nourish a campus garden. Or you noticed a leaky faucet wasting water that could be captured. Identify *one or two opportunities* where a small project could have a big positive impact.

In systems terms, look for places to intervene: *leverage points* (see, “Systems Thinking: A Beginner’s Guide”). A campus example: converting one unused lawn area into a vegetable patch can provide food, habitat, and learning space. In Uzbekistan or Kyrgyzstan, re-planning traditional irrigation channels to slowly feed wetlands can both clean water and support wildlife. Remember Buckton et al.’s qualities for regeneration: diversity, cooperation, and continual learning. How can your idea draw on local diversity (plants, cultures) and build agency for people and nature to thrive together?

Reflection prompts – Spotting opportunities:

- Which of the connected elements feels most in need of change? (E.g. waste overflow, flooding, bland green spaces, idle time). Why?
- Brainstorm three small project ideas that could address that issue. For each, write a one-sentence goal. Which idea would benefit both people and the environment?
- Think about scale: what is the *smallest* viable project you could start (e.g. a single rain barrel, a potted plant experiment, a neighborhood info meeting)? Who else (student clubs, neighbors) might join to amplify it?

Examples and tools:

- Students at **Emory University** (US) saw underused forest edges on campus and launched a student-led rewilding project that restored 700 acres over time with faculty support and grants.
- The **SOS-UK Wilding Campuses** initiative provides case studies of students identifying small but promising plots for ecological restoration. <https://sos-uk.org/wp-content/uploads/l-resources/wilding-campuses-concept-note.pdf>

- **Wageningen University** and **University of Edinburgh** offer open-access materials for identifying regenerative opportunities through circular economy thinking and biodiversity mapping.

4. Design

Plan a concrete action project on the campus or in the community. Make it **small-scale and achievable**, applying your system insights and regenerative principles. Specify: what exactly will you do, who will help, what resources (tools, space, seeds, materials) you need, and what success looks like. Incorporate what you learned in steps 1–3: use local resources, reconnect cycles, and involve others. For example, you could design a **rainwater garden**: collect roof runoff in an on-campus planter to irrigate flowers and vegetables, reducing flood runoff. Or start a **vermicompost bin** in a dorm to turn student scraps into fertilizer for campus trees.

Use the regenerative design mindset: focus on **wholeness and synergy**. In Almaty, for instance, a student team built a project called “*H2O-001*” that filtered water from an irrigation canal to create a fountain supplying drinking water to people, birds and stray animals. They used recycled materials and aimed for minimal waste, embodying sustainability in every choice. Your design can be simpler, but think creatively: perhaps design recycled-art benches, plant native herbs for a pollinator garden, or organize a seed-sharing circle. Plan how you will make the action happen (schedule, roles, communication).

Reflection prompts – Planning your action:

- Outline your project in a few steps. What is the goal? Who does what (e.g. students, staff, community members)?
- How will your design respect natural cycles (e.g. water capture, soil health) and social networks (e.g. student teams, local traditions)?
- Sketch a diagram or moodboard: what will the site look like before and after? How does it fit into the system map you made?
- Identify challenges: what permissions, skills or funds are needed? How might you address them (partner with a club, crowdsource, reuse materials)?

Examples and tools:

- Use the **Nature Positive Universities' “Getting Started” guide** to co-design micro-habitats, native species gardens, or biodiversity corridors.
<https://www.naturepositiveuniversities.net/>
- **Hello World Labs** provides digital tools to plan interventions and visualize potential systems impacts—useful for teams integrating data, community input, and sustainability science. <https://www.helloworlde.com/>
- **Re:wild Your Campus** offers a project starter guide, including steps for engaging with university administration and estimating impact.
<https://www.rewildyourcampus.org/>

5. Learn

Put your project into action *then* step back to reflect and adapt. After implementation, observe the outcomes closely. Collect feedback: talk to others involved and people affected. Measure something if you can (e.g. how much water you collected, pounds of compost produced, number of visitors to a garden). Compare results to your goals. Celebrate what worked and note what didn't.

This reflective step is crucial – it makes the cycle regenerative. As learning theorists remind us, “effective learning occurs through a continuous cycle of experience, reflection, conceptualization, and experimentation”. In other words, each action gives you new data and ideas. Fazey et al. highlight *iterative learning cycles combining action and reflection* as a path to wisdom. Use that: ask “What did we learn?”, “Why?”, and “What will we try next?” For example, if a compost bin isn't breaking down material quickly, you might adjust moisture or mix more leaves. If a social event had low turnout, try a different time or outreach approach.

Reflection prompts – After your action:

- What changed since you started? How did the system respond (environmentally and socially)? Did you see unexpected effects (good or bad)?
- Document the results: take photos or notes. Create a simple report or poster. How close did you come to your goal?
- Plan an iteration: what will you modify or add for the next cycle? Are there new opportunities revealed?
- Share your story with others: write a blog post, social media update or talk with classmates. Spreading what you learned can spark more regenerative ideas.

Examples and tools:

- **Second Nature's implementation framework** includes evaluation rubrics and feedback loops to help track equity and climate resilience progress.
- The **University of Oxford** uses participatory dashboards to track biodiversity and carbon goals across departments, making learning visible and actionable.

Each time you go through Observe→Connect→Identify→Design→Learn, you strengthen your ability to see and support the living system around you. Over the semester or year, this practice can turn small campus projects into bigger cultural shifts. By grounding your work in local context and feedback, you become an agent of regenerative change.

Here's a quick reference list (with links) for students who want to explore or join global networks:

- [Nature Positive Universities](#)
- [Second Nature: Climate Justice in Higher Education](#)
- [Re:wild Your Campus](#)
- [Hello World Labs](#)
- [ThoughtBox Education](#)

- [IARU Green Guide for Universities](#)
- [European University Association Green Deal Roadmap](#)

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Key Terms

- **Regeneration:** Actions or processes that *renew* and *restore* ecosystems and communities. In regenerative systems, “life creates conditions conducive to life” – meaning practices that make environments healthier over time.
- **Systems Thinking:** A way of understanding the world by seeing the *whole system* (not just parts). It emphasizes that “everything is interconnected” and that changes create *feedback loops*. It helps identify how one action affects others.
- **Leverage Point:** A key place in a system where a small change leads to big results. For example, fixing a leaky faucet (small action) can significantly cut water use.
- **Reflexivity:** Deep, ongoing reflection about our own assumptions and actions. In a project, this means continually asking “Why are we doing this, and is it working?” and being ready to change course.
- **Adaptive Cycle (Learning Cycle):** The process of action, reflection, learning, and new action. Kolb’s cycle of *Experience* → *Reflect* → *Think* → *Do* is an example. It ensures projects grow stronger over time.
- **Resilience:** A system’s ability to absorb disturbance and still maintain its core functions. A resilient campus project can handle setbacks (rainstorms, staff changes) and continue benefiting people and nature.